

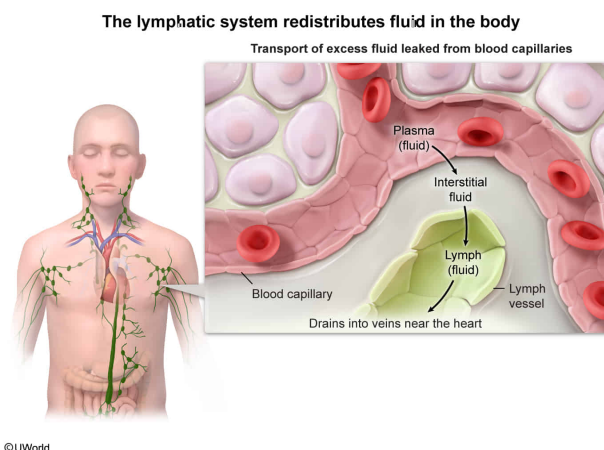
One function of the lymphatic system is to:

- ✔ ☒ A. redistribute fluid in the body. [71%]
- ☐ B. facilitate the exchange of respiratory gases in the lungs. [2%]
- ☐ C. collect filtrate from the blood for excretion. [17%]
- ☐ D. produce plasma proteins. [8%]

Correct

71% answered correctly

Explanation:



As **blood** flows through **blood capillaries**, fluid and small proteins leak from the capillaries into the interstitial space, which contains **interstitial fluid** (ie, extracellular fluid). To maintain proper fluid balance and distribution in the body, this leaked interstitial fluid must be returned by the **lymphatic system**, a network of vessels and nodes that collects and returns excess interstitial fluid to the bloodstream via two large veins near the heart. When fluid collected from the interstitial space is within the vessels and nodes of the lymphatic system, this fluid is referred to as **lymph**.

Therefore, one function of the lymphatic system is to **redistribute fluid in the body**. Additional functions of the lymphatic system include **transport of fats** absorbed by the digestive system to the bloodstream and aiding in the immune response.

(Choice B) The **circulatory** and respiratory (*not* lymphatic) systems are responsible for facilitating the **exchange** of respiratory gases (ie, oxygen and carbon dioxide) in the lungs.

(Choice C) As part of the **urinary** (*not* lymphatic) system, the kidneys **collect filtrate** from the blood for excretion.

(Choice D) The lymphatic system *transports* plasma proteins back to the blood. However, the **liver**, which is *not* part of the lymphatic system, *produces* plasma proteins.

Educational objective:

The lymphatic system is responsible for transporting excess fluid leaked from blood capillaries back to the bloodstream. The lymphatic system is also involved in the immune response and transporting fats absorbed during digestion to the bloodstream.

Time spent:0 sec

Subject:
Biology

QID:404144

System:
Circulation and Respiration